

That's a great idea. Students enjoy assignments much more when they can relate them to real life.

Assignment 3: Searching Algorithms in Real-Life Scenarios

Topics Covered: subArray

Question 1: Netflix Binge Watch Analysis (Sub Array)

Scenario:

Netflix wants to know how many episodes a user watches continuously during a binge session.

Task:

Store the watch time (in minutes) for 7 episodes in an array.

Generate and display all possible continuous watch sessions (subarrays).

Example:

Watch Time = {30, 40, 25}

Subarrays:

{30}
{30,40}
{30,40,25}
{40}
{40,25}

Question 2: Weekly Sales Report (Max & Min)

Scenario:

A shop owner stores daily sales for a week.

Task:

Find:

- Highest Sale
 - Lowest Sale
 - Difference between highest and lowest sale
-

Question 3: Swiggy Delivery Partner Search

Scenario:

Swiggy stores delivery partner IDs.

Task:

- If IDs are unsorted → Use Linear Search.
- If IDs are sorted → Use Binary Search.

Question 4: Stock Market Analysis (Sub Array Challenge)

Scenario:

Daily profits are stored in an array.

Task:

Generate all possible subarrays and calculate:

- Total number of subarrays.
- Largest subarray length.